The shape of a cubic

1

a They become very large in the positive direction

b They become very large in the negative direction

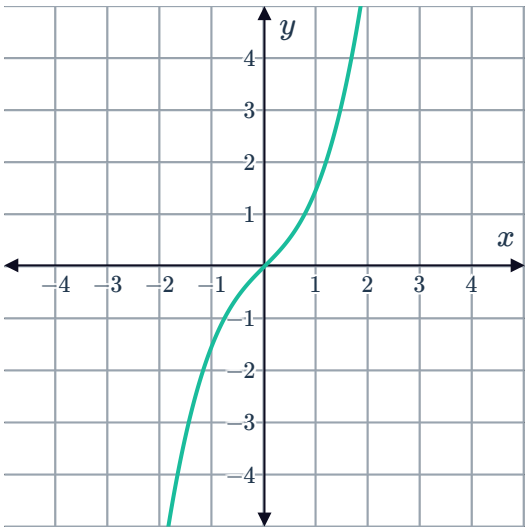
2

a Negative **b** $(0, 3)$ **c** $y = -2x^3 + 3$

3

a Positive

b



c $(0, 0)$

4 $(0, 0)$

5

a C **b** A **c** B

6

a

i $x > 0$ **ii** $x < 0$ **iii** $(0, 3)$

b

i $x > 3$ **ii** $x < 3$ **iii** $(3, 2)$

7

a $x = -2$ **b** $y = 4$

8

a Negative **b** $(0, 3)$ **c** $x < 0$ **d** $x > 0$

e $(0, 3)$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9 $x \geq 3, x = -1$

Transformations

10

a No **b** No **c** No **d** Yes

11

a $(0, 0)$ **b** $(0, 3)$ **c** $(0, 4)$

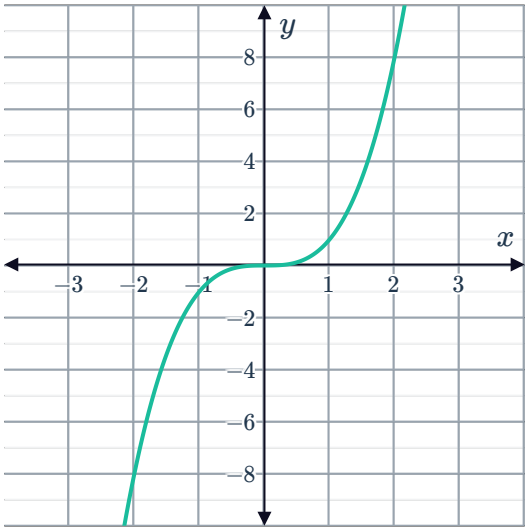
12

a

i

x	-2	-1	0	1	2
y	-8	-1	0	1	8

ii

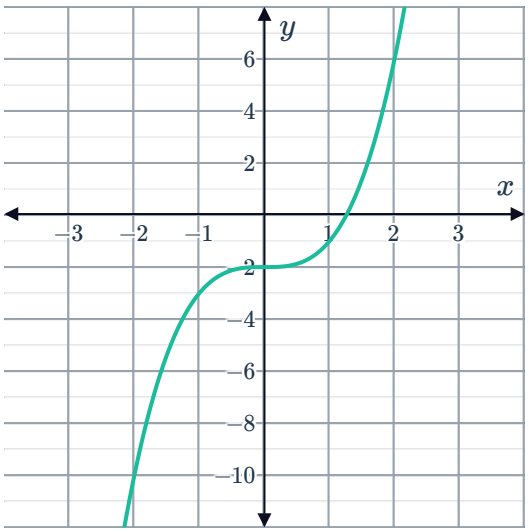


b

i

x	-2	-1	0	1	2
y	-10	-3	-2	-1	6

ii



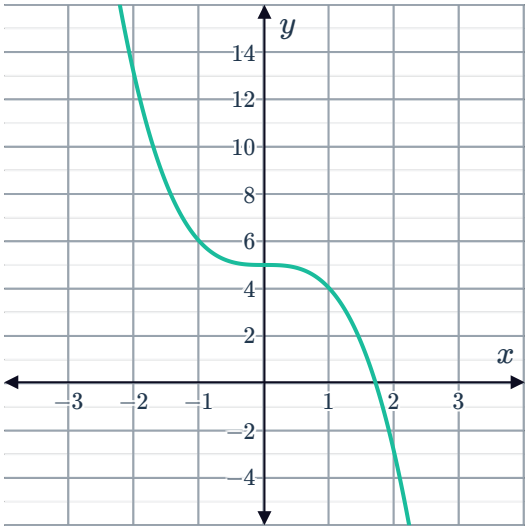
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c

i

x	-2	-1	0	1	2
y	13	6	5	4	-3

ii



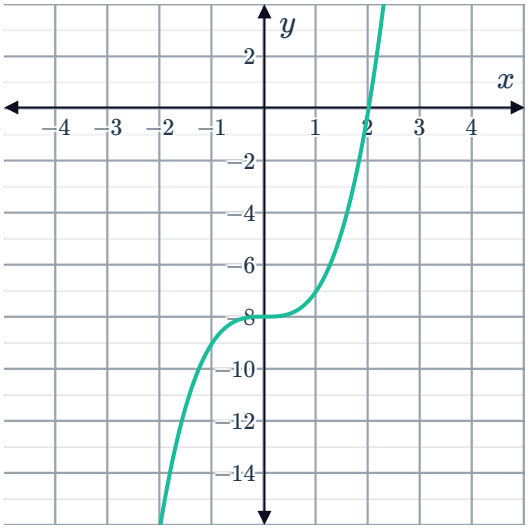
13

a $x = 2$

b $y = -8$

c $(0, -8)$

d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

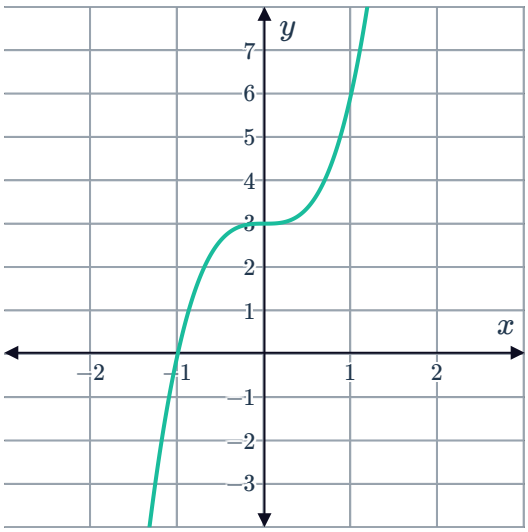
14

a $x = -1$

b $y = 3$

c $(0, 3)$

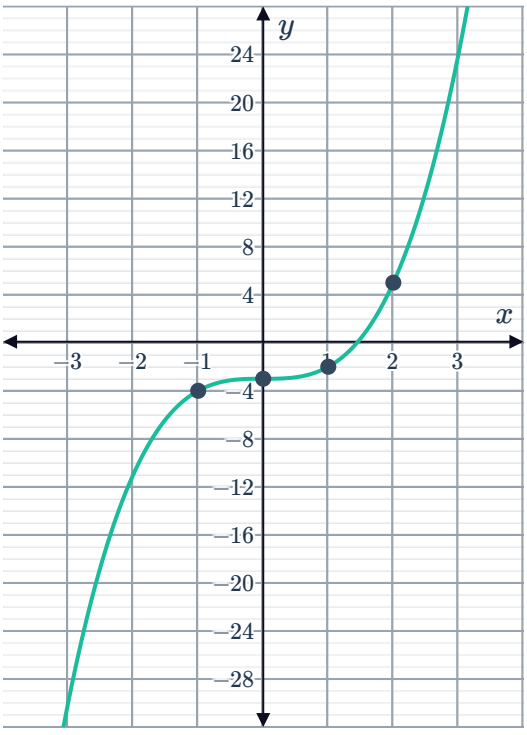
d



15

a $A(-3, -30), B(-2, -11), C(-1, -4), D(0, -3), E(1, -2), F(2, 5), G(3, 24)$

b



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16

a $y = -4$

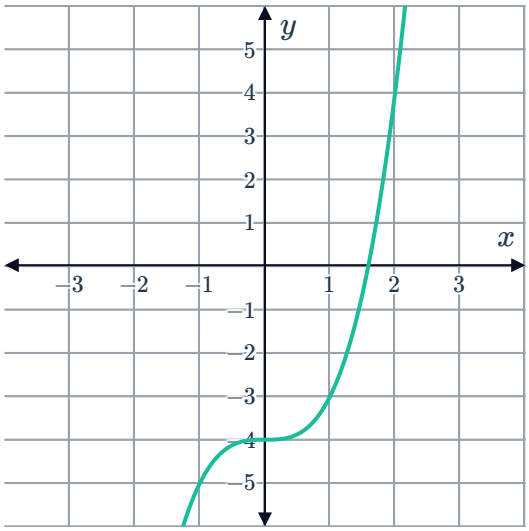
b

x	-1	1	2
y	-5	-3	4

c $(-\infty, \infty)$

d $(-\infty, \infty)$

e



17

a $(2, -4)$

b $(6, -196)$

c



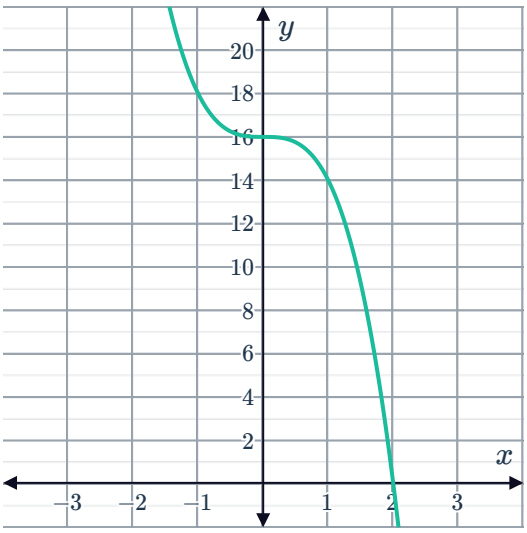
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18

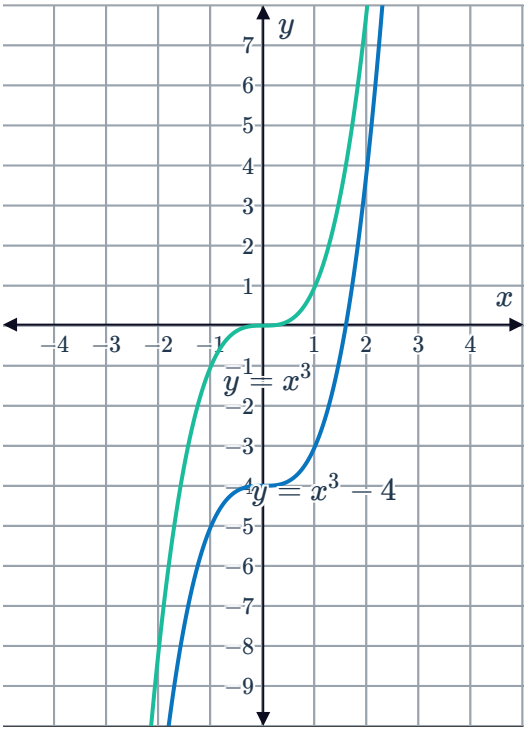
a $x = 2$ **b** $y = 16$ **c** $(0, 16)$ **d** $(-\infty, \infty)$

e $(-\infty, \infty)$

f

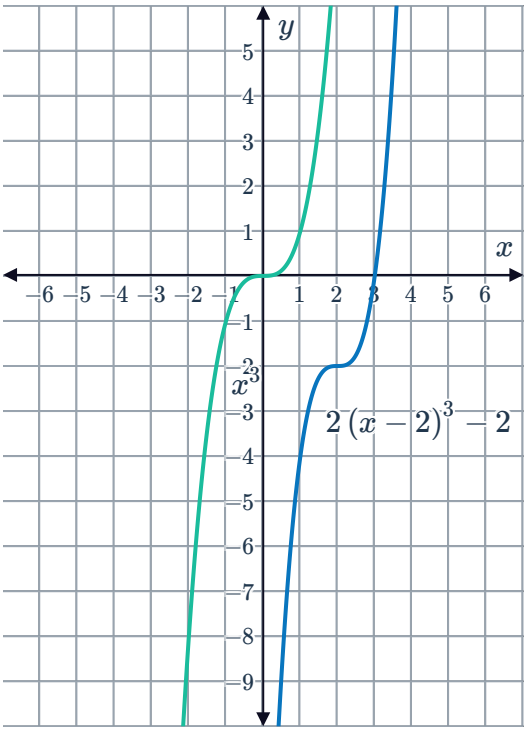


19



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

20



21

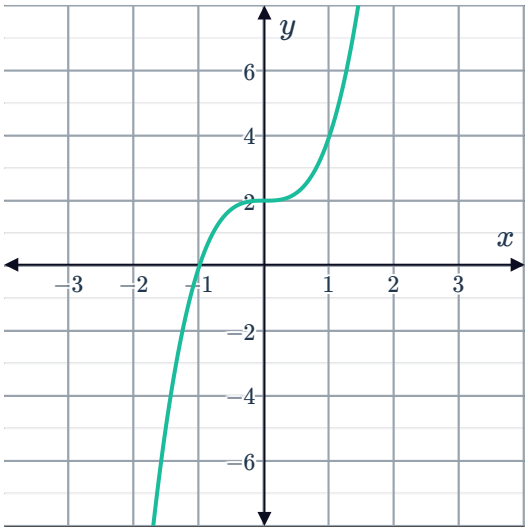
a

i Increasing

ii More steep

iii $(0, 2)$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

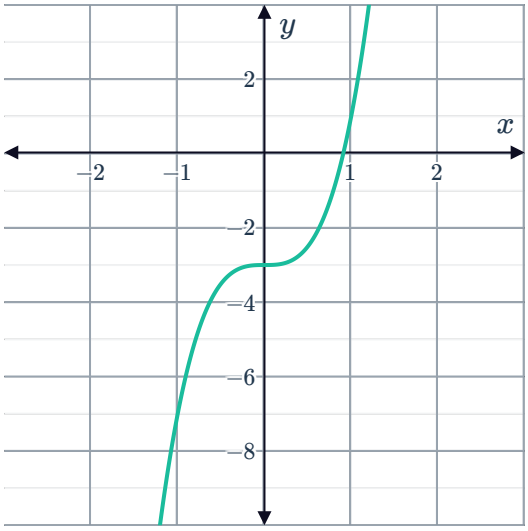
b

i Increasing

ii More steep

iii $(0, -3)$

iv



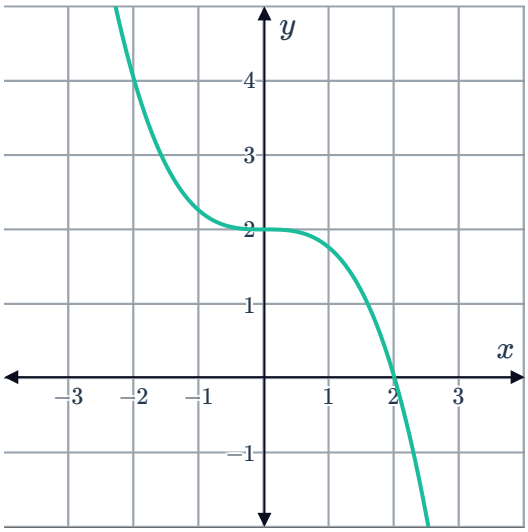
c

i Decreasing

ii Less steep

iii $(0, 2)$

iv

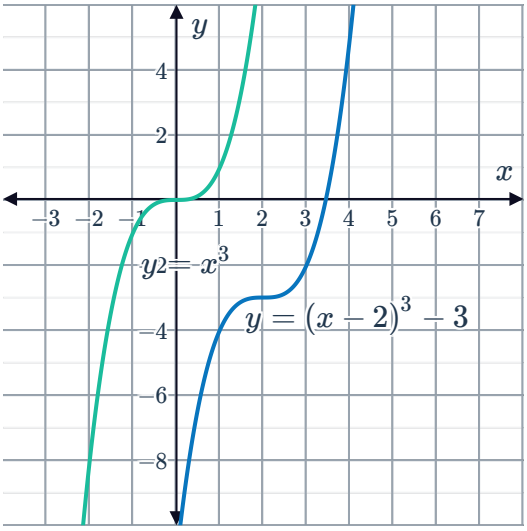


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

22

a Move the graph to the right by 2 units and down by 3 units.

b



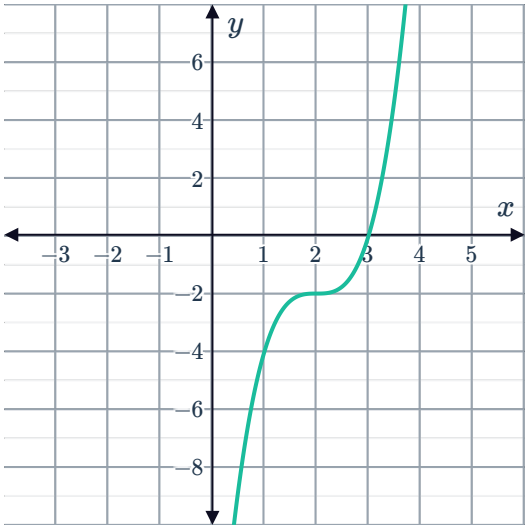
23

a Increasing

b More steep

c $(2, -2)$

d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

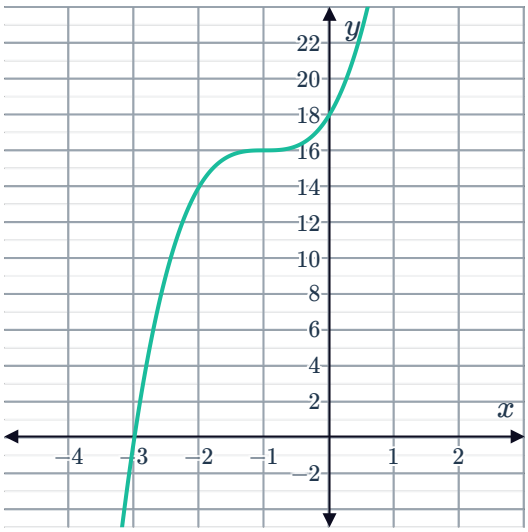
24

a $x = -3$

b $y = 18$

c $(-1, 16)$

d



25

a $x = 2$

b $y = 6$

c $(1, 3)$

d



Factored form of a cubic function

26 3

27 3

28

a No **b** No **c** No **d** Yes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

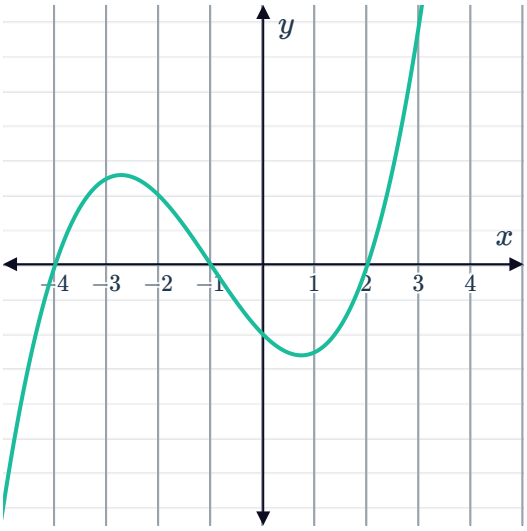
29

a No **b** Yes **c** No **d** No

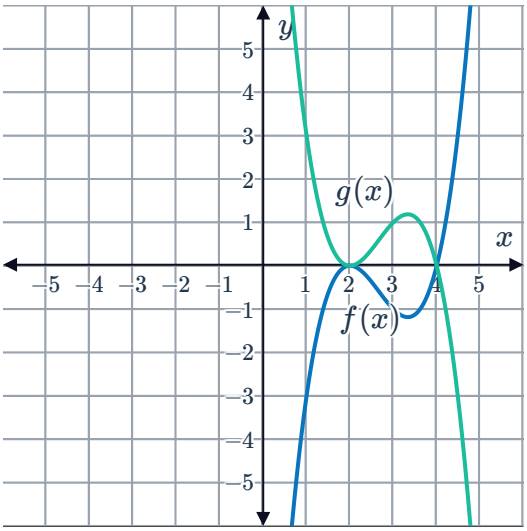
30

a $x = -3, 2, 5$ **b** $y = -(x+3)(x-2)(x-5)$

31



32



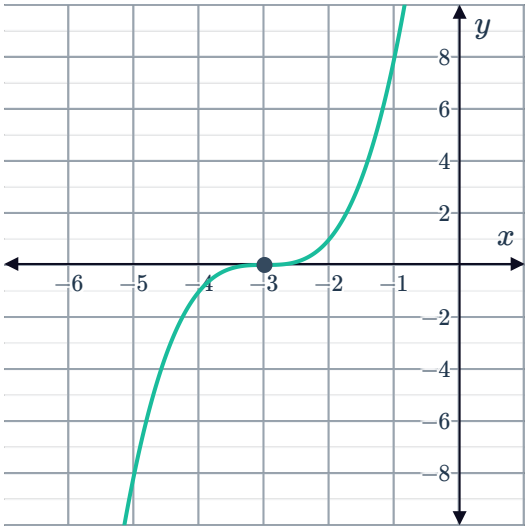
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

33

a

x	-5	-4	-3	-2	-1
y	-8	-1	0	1	8

b



c $(-\infty, \infty)$

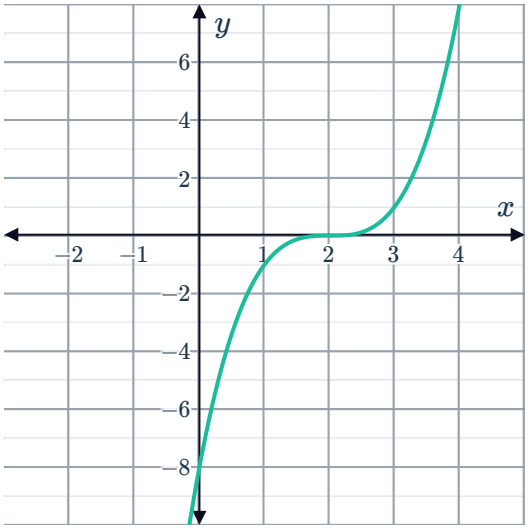
d $(-\infty, \infty)$

34

a

x	0	1	2	3	4
y	-8	-1	0	1	8

b



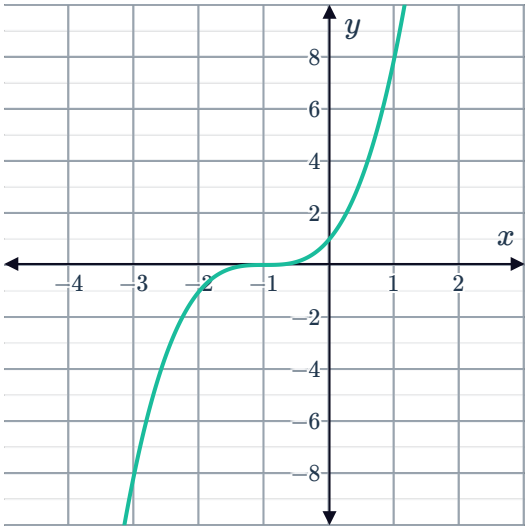
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

35

a

x	-3	-2	-1	0	1
y	-8	-1	0	1	8

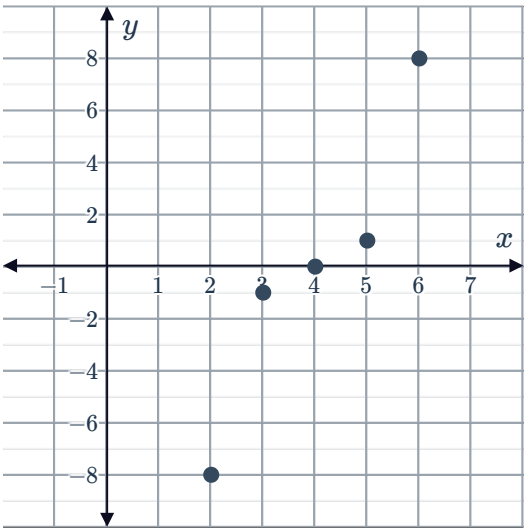
b



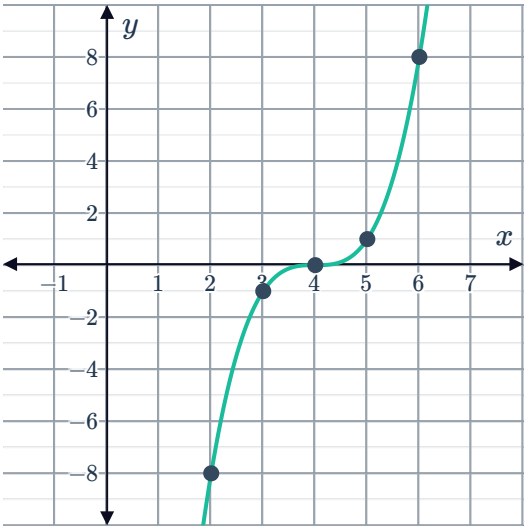
36

a $A(3,-1), B(2,-8), C(5,1), D(4,0), E(6,8)$

b



c



d $8 < y < 27$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

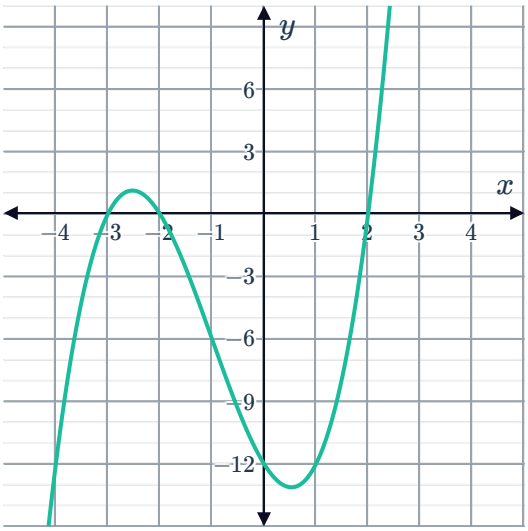
37

a

i $x = -3, -2, 2$

ii $y = -12$

iii

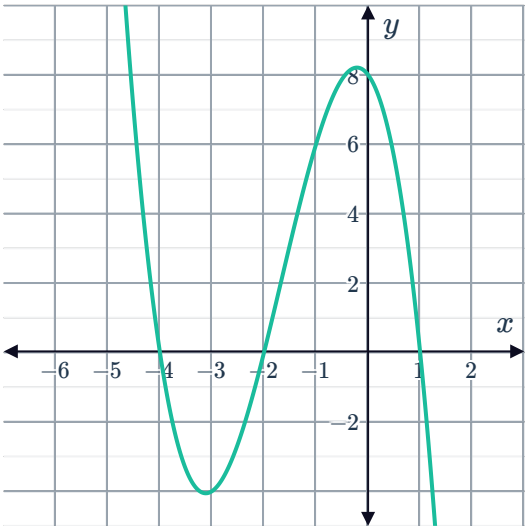


b

i $x = -4, -2, 1$

ii $y = 8$

iii



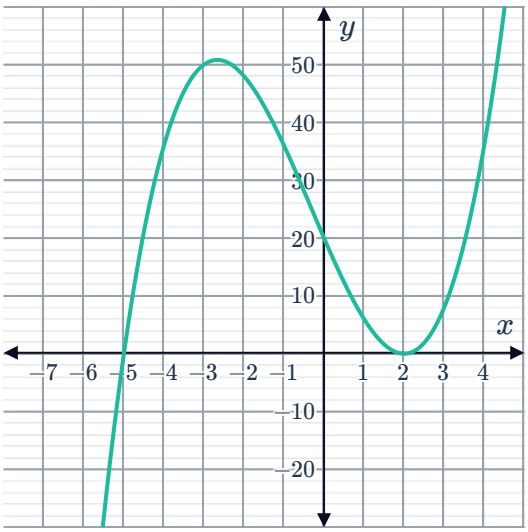
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

c

i $x = 2, x = -5$

ii $y = 20$

iii



38

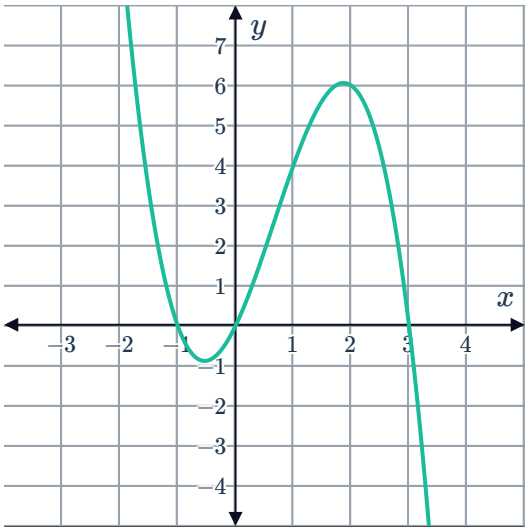
a

i $-x(x-3)(x+1)$

ii $y = 0$

iii $x = 0, 3, -1$

iv



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

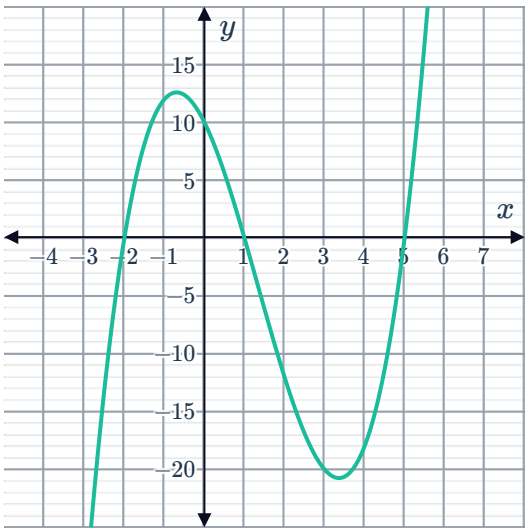
b

i $(x+2)(x-5)(x-1)$

ii $y = 10$

iii $x = -2, 5, 1$

iv



Applications

39

a

r	1	2	4	5	7
V	$\frac{4\pi}{3}$	$\frac{32\pi}{3}$	$\frac{128\pi}{3}$	$\frac{500\pi}{3}$	$\frac{1372\pi}{3}$

b

i No ii Yes iii No iv No

c $r = 6$ m

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

40

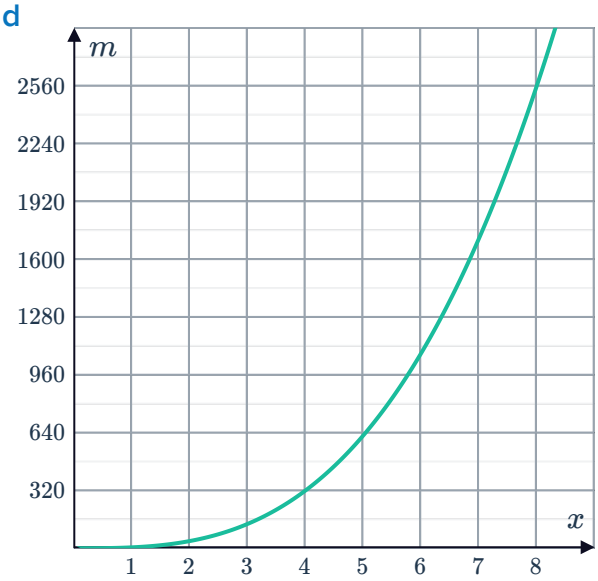
a $k = 5$

b $m = 5x^3$

c

x	1	2	3	4
m	5	40	135	320

d



e 3071 g

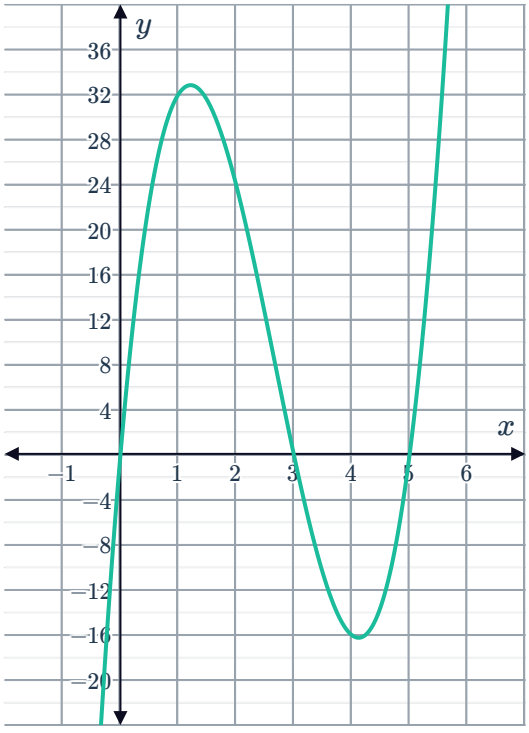
f 7 cm

41

a $V = 4x\,(3 - x)\,(5 - x)$

b $0 < x < 3$

c



d 24 cm^3

42

a $x^2 + h^2 = 15$

b $V = 2\pi h\left(15 - h^2\right)$

c $0 < h < \sqrt{15}$

d $V = 20\sqrt{5}\pi$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647